Inference-Time Scientific Reasoning for Vision-Language Models on mmJEE-Eval: Code Sandboxes, Agentic Correction, Learned Verification and Retrieval, Tested on a Clean Exam Year

Arpit Kumar (12340350) Saurav Gupta (12341940)
Team Erdős, Machine Learning course project, IIT Bhilai

***Abstract*—mmJEE-Eval shows that open vision-language models (VLMs) solve few JEE-Advanced problems and rarely fix their own errors. We wrap a *frozen* 4-bit Qwen3-VL-8B in a training-free framework: a code sandbox for program-of-thought solving, a cross-model Critic/Corrector loop, a learned verifier that selects the final answer, and retrieval of verified worked solutions. Our earlier version (v2) was developed with the 2025 exam as test set. Because 2025 was also used for piloting, and the model may have seen it, we froze an improved version (v3) on 2019–2024 data only and evaluated both versions on a *clean* test set: JEE-Advanced 2026 (204 questions, English and Hindi), which postdates the model. On 2026 the v3 framework reaches 58.8% (95% CI [50.0, 67.2]) vs. 44.3% for chain-of-thought, and beats self-consistency at the same generated-token budget by +6.4 points (CI [+1.5, +10.8], McNemar p=0.011). The frozen v2 framework reaches 55.4% and also beats matched self-consistency (+6.4). So the central claim holds on unseen data. v3 improves the solver (+6.8 points for PoT), but its overall advantage over v2 (+3.4) is not significant. On 2026 the new verifier selects no better than majority voting over the same pool, so the gain comes from a better candidate pool.**

I. Introduction

JEE-Advanced questions combine a diagram, multi-step physical or chemical reasoning, and exact answers. mmJEE-Eval [1] turns them into an image-only, bilingual benchmark and reports two weaknesses of open VLMs: low accuracy, and single-pass self-correction that turns a detected error into a correct answer only 1.1–5.2% of the time.

Our SOP proposed to attack both at inference time, without fine-tuning. The v2 system did so on the 2025 exam. Two threats remained. First, 2025 was used to pilot the system, so design choices may fit it. Second, Qwen3-VL may have seen the 2025 papers. This report addresses both:

- A **clean test year**. JEE-Advanced 2026 postdates the model. Its English questions are in the dataset; we built the Hindi twins ourselves from the official papers.
- A **pre-registered freeze**. Every v3 change was chosen on 2019–2024 data, by leave-one-year-out (LOYO) cross-validation or on the 2024 dev year, with inclusion rules fixed in advance. It was then frozen (`V3_FREEZE.md`) before any 2026 or 2025 run.
- **Both frameworks on 2026**. The frozen v2 system is re-run unchanged next to v3, with compute-matched self-consistency (SC) [2], cluster-bootstrap CIs and McNemar tests for every claim.
- **Analyses**: correction dynamics in the paper's terms, a learned correction gate, the verifier's distribution shift, a bridge to the paper's Qwen2.5-VL-7B, and memorisation probes.

II. Related work

Tool-augmented reasoning. Program-of-thought [3] and PAL [4] move arithmetic into an interpreter. **Sampling and voting.** Self-consistency [2] votes over sampled chains and is our compute-matched baseline. **Self-correction.** Self-Refine [5] and CRITIC [6] iterate critique and revision, but intrinsic self-correction often fails without external signal [7]. We add code execution, step localisation and a cross-model critic. **Verifiers.** Outcome and process verifiers [8], [9] rank sampled solutions; ours is a light gradient-boosted ranker trained on automatically labelled training-year pools. **Retrieval.** RAG [10] supplies context. We retrieve verified worked solutions with FAISS [11] and bge [12] embeddings.

III. Data and evaluation protocol

Splits. 2019–2024 (1,270 questions) are used for training and development; 2024 is the dev year. **Clean test:** 2026, 204 questions (102 English from the dataset plus their 102 Hindi twins). We cropped the Hindi twins from the official Hindi papers, matched them by paper, subject and number, took answers from the English key. Automatic checks match every twin, and 14 crops, including every size outlier, were checked by hand against their English twins. Their style differs from the dataset's images, so results are also reported per language. **Post-hoc set:** 2025, 190 questions. It was the v2 test set and was used to pilot v1/v2, so it is reported separately and labelled post-hoc.

Scoring and metrics. We use the paper's `extract_answer` and `is_answer_correct`, copied verbatim and checked by tests. Pass@1 is the mean over samples. Selected-answer rows report the accuracy of the selected answer. CIs use a cluster bootstrap over EN/HI twin pairs (2,000 resamples), and differences use the paired version. McNemar treats questions as independent, so its

TABLE I
DEV A/B TESTS ON 2024 (204 QUESTIONS) THAT DECIDED THE FREEZE. RULE FIXED IN ADVANCE: INCLUDE IFF $\Delta > 0$ AND THE CI LOWER BOUND > 0.

Arm	Base	Arm	Δ [95% CI]
Transcript block (PoT)	50.2	55.1	+4.9 [+1.8, +8.2]
Adaptive budget (PoT)	50.2	52.6	+2.3 [+1.0, +3.7]
Adaptive budget (CoT)	48.5	54.9	+6.4 [+2.5, +10.8]
RAG, v2 recipe	50.2	50.4	+0.2 [−2.3, +2.9]
RAG v3 (hybrid + JEEBench)	50.2	50.6	+0.3 [−2.1, +3.0]

p is optimistic. **Compute matching:** SC uses the smallest number of CoT samples whose generated tokens reach the framework's (capped at the 20 samples drawn).

Leakage control. No test question or twin is in any training pool. KB entries close to any 2025 or 2026 question are removed (de-duplication against both years). The v3 freeze used no 2025 or 2026 data.

IV. METHOD

Shared core (v2). *A. Code sandbox:* the solver writes numbered steps and puts every calculation in a `python` block. The block runs in an isolated subprocess (AST whitelist, no I/O, resource limits), and its output is fed back before generation continues. If no `\boxed{}` answer appears, a plain-CoT fallback answers. *B. Correction:* a cross-model critic (InternVL3.5-8B [15]) names the first wrong step or says `NO_ERROR`. The corrector keeps the earlier steps and re-derives from the flagged one, for up to 3 rounds; it stops early when the critic accepts the solution or the answer is unchanged for two rounds. *C. Verifier:* a classifier over handcrafted and embedding features picks one answer from 16 candidates (8 originals + 8 corrected). *D. Retrieval:* a one-line concept tag retrieves up to two verified worked solutions from 2019–2024 as exemplars.

v3 changes (Table I): (i) *Adaptive budget.* A cut-off answer is continued once for 4,096 more tokens. This applies to every row, including the baselines, so comparisons stay fair. (ii) *Transcript block.* The solver prompt also contains the VLM's own English transcription of the question (Hindi is translated), to help it read. (iii) *RAG v3.* Hybrid tag + problem-text similarity, a KB extended with JEEBench 2016–2018, and τ tuned by LOYO. LOYO coverage is 11.7% vs. 8.4% for the v2 recipe at the same proxy precision. (iv) *Verifier v3.* An XGBoost pairwise ranker, selected by LOYO over 2019–2024. It is trained on pools that, like the test pools, contain corrected solutions, and it gets critic features (flagged, rounds, answer changed) and agreement over the whole pool and its halves. LOYO pick accuracy is 62.0% vs. 53.7% for majority voting (oracle 78.5%). (v) *Correction gate.* A chain keeps its corrected answer only if the verifier score rises by more than $\delta = -0.08$, with δ chosen out-of-year. This is reported as its own row; the framework's final answer is still the verifier's pick from the full pool.

Setup. Qwen3-VL-8B-Instruct [14] and InternVL3.5-8B (AWQ 4-bit) run with vLLM [13] on one RTX 3060, one model at a time. Sampling uses $T=0.7$, top-p 0.8 and top-k 20, the same for all rows. CoT takes 20 samples per question; PoT and correction take 8. Every call is cached in SQLite and seeds are deterministic, so the three lab shutdowns during the final batch resumed without changing any result.

V. RESULTS ON THE CLEAN 2026 TEST

A. Main result

Table II and Fig. 2 give the SOP's Table I for both frameworks. The v3 framework reaches **58.8%** vs. 44.3% for single-pass CoT (+14.6 points). At 80k generated tokens per question, matched SC uses 18 samples and reaches 52.5%. The framework is +6.4 points better (CI [+1.5, +10.8]): it answers 18 questions that SC misses, against 5 the other way. The frozen v2 framework reaches 55.4%, also +6.4 points above its matched SC (CI [+1.5, +11.8], $p=0.015$). *The main v2 claim, beating compute-matched self-consistency, therefore holds on an exam the model cannot have seen.* Fig. 3 shows that every framework row lies above the SC curve at its cost.

B. v3 vs. v2

Table III gives the paired tests. The v3 solver is clearly better: v3 PoT beats v2 PoT by +6.8 points (CI [+3.6, +10.0]). The adaptive budget alone lifts CoT by +3.0. The full v3 framework is +3.4 points above v2 (CI [−1.5, +8.8], $p=0.230$), which is *not* significant on 204 questions. On the post-hoc 2025 set the same comparison is +6.8 (CI [+1.1, +12.1]). The difference between the two years is largely the verifier (Sec. V-E).

C. Where the frameworks win and lose

Table IV shows that v3 improves on most groups, but the gaps the paper reports remain. For v3 on 2026, English reaches 63.7% and Hindi 53.9%; text-only questions reach 72.5% and diagram questions only 30.3%. Inference-time reasoning does not fix perception. Part of the Hindi gap on 2026 may come from our own crops.

D. Correction loop and the correction gate

In the paper's terms (Table V), the cross-model critic flags 52.9% of wrong v3 solutions in round 1 (69.0% for v2). Round-1 correction fixes 13.2% of the detected errors (10.8% for v2), well above the paper's 1.1–5.2% single-pass rate. The critic also flags 19.9% of *correct* solutions. Because the v3 solver is stronger, fewer wrong solutions remain, and the correction loop gains less in absolute terms.

The gate (Table VI) does what it was trained for. On 2026 it cuts right→wrong flips from 33 to 9 while keeping most fixes (68→54 wrong→right). Per-chain accuracy goes from 52.0% to 52.6% (+0.6, CI [−0.4, +1.5]); the change is small and not significant. The flip pattern carries over from the training years to both test years.

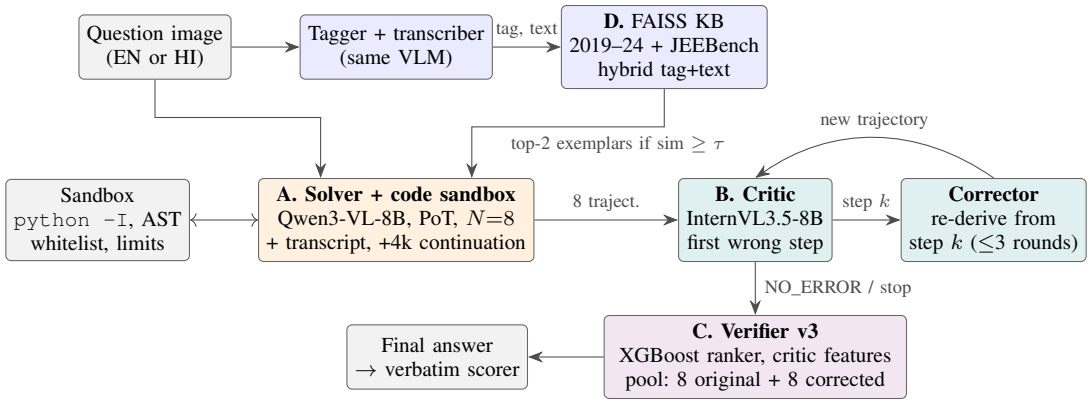


Fig. 1. The v3 framework around a frozen VLM. Only the verifier (C) is trained, on automatically labelled 2019–2024 pools. The v3 changes from v2 are the transcript, the continuation, hybrid retrieval with JEEBench, and the verifier v3 with its correction gate.

TABLE II

TABLE I ON THE CLEAN 2026 TEST SET (204 QUESTIONS). TOP: FROZEN V3 FRAMEWORK. BOTTOM: THE FROZEN V2 FRAMEWORK RE-RUN UNCHANGED. ACCURACY IN %; 95% CLUSTER-BOOTSTRAP CIS OVER EN/HI TWINS; TOKENS ARE GENERATED TOKENS.

Configuration	Acc.	Num.	95% CI	Δ vs. baseline [CI]	Tok./q	Calls/q
v3 framework						
Baseline (single-pass CoT, adaptive budget)	44.3	29.5	37.4–51.0	–	4.5k	1.0
Self-consistency (matched compute) (n=18)	52.5	34.6	43.1–61.3	+8.2 [+4.5, +12.0]	80.4k	18.0
+ Code sandbox + RAG v3 + transcript (v3 PoT)	49.8	37.3	42.5–56.9	+5.6 [+2.0, +9.2]	6.5k	2.2
+ Learned verifier v3 (PoT pool only)	57.8	41.0	48.5–66.2	+13.6 [+8.4, +18.6]	51.8k	18.3
+ Correction (cross-model critic)	52.0	39.6	44.6–59.2	+7.7 [+4.3, +11.0]	10.0k	5.1
+ Correction (gated)	52.6	39.9	44.9–60.1	+8.3 [+4.7, +11.7]	10.0k	5.1
v3 framework (verifier v3)	**58.8**	44.9	50.0–67.2	+14.6 [+9.6, +19.5]	80.3k	41.7
v2 framework (re-run)						
Baseline (single-pass CoT)	41.2	26.5	34.4–47.9	–	2.9k	1.0
Self-consistency (matched compute) (n=20, needs 21)	49.0	28.2	40.2–57.4	+7.8 [+4.4, +11.3]	58.0k	20.0
+ Code sandbox	43.0	31.1	36.4–49.6	+1.8 [−1.1, +4.7]	4.0k	1.8
+ Agentic correction (cross-model critic)	46.8	36.2	40.1–53.4	+5.5 [+2.1, +9.0]	7.4k	4.7
+ Learned verifier	55.4	39.7	46.6–63.7	+14.2 [+9.0, +19.5]	59.2k	38.0
Full framework	**55.4**	39.7	46.6–63.7	+14.2 [+9.0, +19.3]	60.1k	38.8

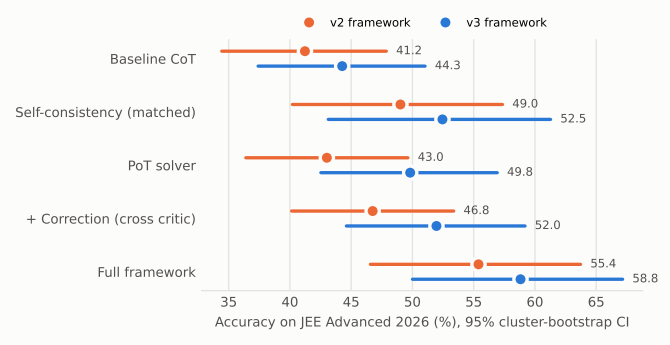


Fig. 2. 2026 clean test: matching rows of the v2 and v3 frameworks, with 95% cluster-bootstrap CIs.

TABLE III

PAIRED COMPARISONS (CLUSTER BOOTSTRAP; MCNEMAR COUNTS QUESTIONS ONLY a / ONLY b GETS RIGHT; IT IS MEANINGFUL FOR SINGLE-ANSWER ROWS ONLY, AS PASS@1 ROWS ARE ROUNDED PER QUESTION). †POST-HOC.

Comparison (a vs. b)	Δ [95% CI]	P(Δ≤0)	a/b	p
2026 v3 fw vs. v2 fw	+3.4 [−1.5, +8.8]	0.113	16/9	0.230
2026 v3 PoT vs. v2 PoT	+6.8 [+3.6, +10.0]	0.000	27/3	<0.001
2026 CoT adaptive budget vs. CoT	+3.0 [+1.7, +4.5]	0.000	6/3	0.508
2026 gated vs. ungated correction (v3)	+0.6 [−0.4, +1.5]	0.122	4/1	0.375
2026 verifier v3 pick vs. v3 PoT mean	+9.0 [+5.8, +12.4]	0.000	17/0	<0.001
2025† v3 fw vs. v2 fw	+6.8 [+1.1, +12.1]	0.012	21/8	0.024
2025† v3 PoT vs. v2 PoT	+8.1 [+4.8, +11.6]	0.000	30/7	<0.001

E. Verifier: classifier vs. selector

As a classifier, verifier v3 transfers well: its AUC on the 2026 pool is 0.90. *As a selector, on the clean year, it adds nothing*: it picks 58.8% vs. 59.3% for majority voting over the same 16 answers (−0.5), with an oracle of 79.4%. On 2025 it beats majority by +6.3 points. So the v3 gain on 2026 comes from a better candidate pool (transcript, continuation, correction), not from selection. Picking among the 8 PoT samples alone already reaches 57.8%.

The shift stated before the runs is visible in Table VIII. The verifier was trained on plain PoT solutions of about 3.5k tokens (37.2% cut off). The 2026 v3 solutions are 6.5k long because of the transcript and continuation, and 40.3% are cut off. This is a plausible reason the selector gain does not carry

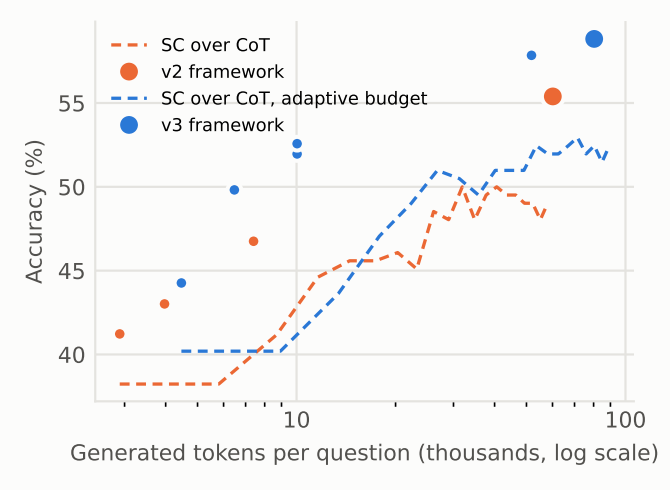


Fig. 3. 2026: accuracy vs. generated tokens. Dashed: SC over n CoT samples (majority vote). Dots: Table I rows; large dots: the full frameworks.

TABLE IV
FULL-FRAMEWORK ACCURACY (%) BY GROUP. 2025 IS POST-HOC.

Group	v2 '26	v3 '26	v2 '25	v3 '25
English	60.8	63.7	73.7	74.7
Hindi	50.0	53.9	56.8	69.5
Text only	69.6	72.5	72.5	81.9
Needs image	25.8	30.3	46.2	46.2
Physics	32.4	35.3	54.8	64.5
Chemistry	52.9	60.3	60.9	59.4
Mathematics	80.9	80.9	79.7	92.2
MCQ-Single	79.2	83.3	67.4	65.2
MCQ-Multiple	55.6	55.6	59.5	69.0
Numerical	39.7	44.9	69.0	77.4
Matching	58.3	62.5	55.6	72.2

TABLE V
CORRECTION LOOP, PAPER-STYLE METACOGNITION (%). EP: WRONG SOLUTIONS FLAGGED IN ROUND 1; EC: ROUND-1 FIXES AMONG DETECTED; FA: CORRECT SOLUTIONS FLAGGED. W→R / R→W: ANSWER FLIPS AFTER ALL ROUNDS.

Run	Chains	Acc.	EP	EC	FA	W→R	R→W
2026 pot_corr_cross	1632	43.0→46.8	67.5	9.7	23.6	76	15
2026 full	1632	44.7→48.8	69.0	10.8	25.2	83	16
2026 v3_full	1632	49.8→52.0	52.9	13.2	19.9	68	33
2025 v3_full	1520	56.2→57.9	40.3	13.1	23.3	46	21
2025 pot_corr_cross (v2)	1520	48.2→51.3	59.3	13.3	25.7	87	39
2025 full (v2)	1520	50.9→54.7	60.7	15.7	28.6	87	29

TABLE VI
CORRECTION GATE: PER-CHAIN ACCURACY (%) AND FLIPS, UNGATED → GATED.

Set	Chains	Orig.	Corr.	Gated	W→R	R→W
Train (LOYO, picks δ)	1936	44.4	47.6	**48.2**	95→88	32→14
2026 (clean)	1632	49.8	52.0	**52.6**	68→54	33→9
2025 (post-hoc)	1520	56.2	57.9	**58.5**	46→41	21→7

over to the clean year.

TABLE VII
PRIMARY VERIFIER ON ITS OWN 16-ANSWER TEST POOL: AUC AND ACCURACY OF THE PICKED ANSWER VS. POOL BASELINES (%).

Pool	Verifier	AUC	Pick	Maj.	Rand.	Oracle
2026 v3	xgb_ranker/handcrafted	0.902	**58.8**	59.3	50.9	79.4
2026 v2	logreg/all	0.913	**55.4**	53.9	46.7	76.5
2025 v3	xgb_ranker/handcrafted	0.872	**72.1**	65.8	57.1	87.4
2025 v2	logreg/all	0.836	**65.3**	65.8	52.8	85.3

TABLE VIII
SHIFT BETWEEN THE VERIFIER'S TRAINING SOLUTIONS AND THE TEST POOLS. CUT: FINISH = LENGTH ON THE FINAL CALL.

Solutions	Rows	Acc.	Cut	Tok.	Acc. cut	Acc. compl.
train 2019-2024 pot_train	10160	45.5	37.2	3.5k	28.9	55.3
2026 pot	1632	43.0	42.8	4.0k	27.0	55.0
2026 v3_pot	1632	49.8	40.3	6.5k	32.0	61.8
2025 pot	1520	48.2	36.7	3.5k	29.9	58.7
2025 v3_pot	1520	56.2	36.6	5.9k	41.7	64.7

VI. POST-HOC 2025, MODEL BRIDGE AND MEMORISATION

2025 (post-hoc). The v3 framework reaches 72.1% (Table IX) vs. 65.3% for the frozen v2 run, and beats matched SC by +13.7 points. Since 2025 shaped earlier design choices and may be in the model's training data, we do not base claims on it. The larger gaps on 2025 than on 2026 are consistent with some optimism on that year.

Bridge to the paper's model. With Qwen2.5-VL-7B (2025 is clean for this model), the framework does not help: 14.0% for CoT, 16.3% for matched SC and 15.3% for the framework (-1.1 vs. SC, CI $[-7.4, +5.3]$; Table X). At about 14% accuracy, most candidates are wrong, so agreement, critique and verification carry little signal. The framework needs a solver that is right often enough.

Memorisation probes. Asked for the answer only, Qwen3-VL scores 10.5–15.7% on 2019–2024, 13.2% on 2025 and 15.2% on the clean 2026 (Table XI). The clean year is not lower, so we see no memorisation signal. Pass@5 of v3 PoT on 2026 is 70.2%, well above any selector, so there is still headroom in the pool.

VII. DISCUSSION

What holds on clean data. Code-based sampling, cross-model correction and selection beat voting on plain CoT at the same cost, for both v2 and v3. The transcript block and the adaptive budget give a significantly stronger solver.

What does not. Verifier v3 beat majority voting under LOYO and on 2025, but not on 2026. Its features shift with the longer v3 solutions, and its selector gain does not survive the change. The correction gate removes most harmful flips but barely moves accuracy. Retrieval stays a small effect.

Remaining failures. Diagram questions and Hindi questions still lag far behind, and about two in five PoT trajectories still end cut off.

TABLE IX
V3 TABLE I ON 2025 (190 QUESTIONS, **POST-HOC**: 2025 WAS USED TO PILOT V1/V2).

Configuration	Acc.	Num.	95% CI	Δ vs. baseline [CI]	Tok./q	Calls/q
Baseline (single-pass CoT, adaptive budget)	48.3	43.2	41.7–55.3	–	4.0k	1.0
Self-consistency (matched compute) (n=18)	58.4	52.4	50.0–66.8	+10.1 [+6.1, +14.1]	72.4k	18.0
+ Code sandbox + RAG v3 + transcript (v3 PoT)	56.2	59.7	49.5–63.0	+8.0 [+3.3, +13.1]	5.9k	2.3
+ Learned verifier v3 (PoT pool only)	68.9	72.6	61.1–76.8	+20.7 [+14.7, +27.3]	47.3k	19.5
+ Correction (cross-model critic)	57.9	62.5	51.1–64.5	+9.6 [+4.8, +15.0]	8.7k	5.0
+ Correction (gated)	58.5	63.7	51.7–65.1	+10.2 [+5.3, +15.6]	8.7k	5.0
v3 framework (verifier v3)	**72.1**	77.4	64.7–80.5	+23.8 [+17.4, +30.7]	69.9k	41.1

TABLE X
BRIDGE: THE FRAMEWORK WITH THE PAPER'S QWEN2.5-VL-7B ON 2025 (CLEAN FOR THAT MODEL; VERIFIER TRANSFERRED FROM QWEN3).

Configuration	Acc.	Num.	95% CI	Δ vs. baseline [CI]	Tok./q	Calls/q
Baseline (single-pass CoT)	14.0	6.6	11.0–17.3	–	0.7k	1.0
Self-consistency (matched compute) (n=20, needs 33)	16.3	10.7	10.0–23.2	+2.3 [−1.9, +6.5]	14.1k	20.0
+ Code sandbox	13.9	5.7	10.2–18.0	−0.1 [−2.9, +2.9]	1.0k	3.6
+ Correction (same-model critic)	13.9	6.1	10.2–17.9	−0.1 [−2.9, +2.9]	1.5k	6.7
+ Correction (cross-model critic)	13.7	8.0	10.3–17.3	−0.3 [−2.9, +2.5]	2.8k	11.6
+ Learned verifier (transfer)	15.3	8.3	10.0–21.1	+1.3 [−3.3, +6.0]	22.7k	92.9

TABLE XI
MEMORISATION PROBE: ANSWER-ONLY ACCURACY (%, 64 TOKENS) BY EXAM YEAR. 2026 IS AFTER BOTH MODELS' RELEASE.

Model	'19	'20	'21	'22	'23	'24	'25	'26
Qwen3-VL-8B	12.5	14.6	11.8	10.5	14.5	15.7	13.2	15.2
Qwen2.5-VL-7B	12.5	12.3	14.5	17.1	13.5	14.2	12.1	21.1

VIII. LIMITATIONS

One substrate model; one clean year of 204 questions (CIs about ±8 points), so differences of a few points are not resolvable. The Hindi 2026 crops are our own. The transcript and the KB texts come from the same VLM. Compute matching counts generated tokens, not prompt tokens or code execution. The verifier was not retrained for the v3 distribution.

IX. CONCLUSION

On an exam year the model cannot have seen, a frozen 8B VLM with a code sandbox, cross-model correction and answer selection improves accuracy from 44.3% to 58.8% and beats compute-matched self-consistency by +6.4 points. The v2 framework's claim reproduces on the clean year (+6.4). v3's pre-registered changes make the solver stronger, but its total advantage over v2 is not significant, and its verifier does not beat majority voting there. The next steps are a selector robust to solution-length shift and better perception for diagrams and Hindi.

REPRODUCIBILITY

Code, configs and tests are in `mmjee-reasoner/`. The frozen v3 config is `configs/v3_final.yaml` (`V3_FREEZE.md`). Evaluation uses `configs/v3_eval.yaml` and `configs/v3_eval_v2fw.yaml`, and `scripts/final_report_v3.py` regenerates every table, macro and figure of this report.

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